THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. From the top of a tower $6\sqrt{3}$ m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 30° and 60°. The distance between the two objects is: 1

(a) 24 m **(b)** $12\sqrt{3}$ m **(c)** 6 m **(d)** 12 m

2. Two concentric circles have radii 68 cm and 32 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 60 cm **(b)** 120 cm **(c)** 72 cm **(d)** 100 cm

3. A sector of a circle of radius 42 cm has central angle 45°. The perimeter of the sector is: 1

(a) 75 cm **(b)** 117 cm **(c)** 159 cm **(d)** 33 cm

4. A solid cone and a solid hemisphere have equal base radii 7 cm and equal volumes. The height of the cone is: 1

(a) 10.5 cm **(b)** 14 cm **(c)** 21 cm **(d)** 7 cm

5. For the data with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 with frequencies 3, 13, 11, 17, 15, 24, 17 respectively, The median of the data is: 1

(a) 120 **(b)** 108 **(c)** 100 **(d)** 103.6

6. A bag contains 7 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{4}{7}$. The value of x is: 1

(a) 4 **(b)** 10 **(c)** 5 **(d)** 6

7. Which of these is a rational number? 1

(a) $\sqrt{2}+\sqrt{3}$ **(b)** $2\sqrt{5}$ **(c)** $\dfrac{\sqrt{12}}{\sqrt{3}}$ **(d)** $\sqrt{4}$

8. For the cubic with zeroes $1, 2, 3$, $\alpha\beta+\beta\gamma+\gamma\alpha$ is: 1

(a) 36 **(b)** 11 **(c)** -11 **(d)** 6

9. If $x+y=7$ and $xy=10$, then $x^2+y^2=$ 1

(a) 29 **(b)** 49 **(c)** 39 **(d)** 69

10. The equation $kx^2+6x+3=0$ has real roots when: 1

(a) $k \geq 3$ **(b)** $k < 3$ only

(c) $k \leq 3$ and $k \neq 0$ **(d)** $k > 0$

11. The sum of the series $5+9+13+\cdots+101$ is: 1

(a) 1325 **(b)** 1430 **(c)** 2650 **(d)** 1224

12. In $\triangle ABC$ right-angled at B, $AB=3$, $BC=4$. If $BD \perp AC$, then AD (in cm) is: 1

(a) $\frac{16}{5}$ **(b)** $\frac{3}{2}$ **(c)** $\frac{9}{5}$ **(d)** $\frac{12}{5}$

13. The side of the square with diagonal endpoints $(0,0)$ and $(6,8)$ has squared length: 1

(a) 25 **(b)** 200 **(c)** 100 **(d)** 50

14. $\dfrac{\sin^2 25^\circ + \sin^2 65^\circ}{\cos^2 35^\circ + \cos^2 55^\circ} =$ 1

(a) $\frac{1}{2}$ **(b)** 2 **(c)** 0 **(d)** 1

15. From the top of a building 3 m high, the angle of elevation of the top of a tower is 30° and the angle of depression of its foot is 60°. The height of the tower is: 1

(a) 4 m **(b)** $4\sqrt{3}$ m **(c)** $\sqrt{3}$ m **(d)** 1 m

16. In $\triangle ABC$, right-angled at B, $AB = 45$ cm and $BC = 60$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 30 cm **(b)** 15 cm **(c)** 16 cm **(d)** 37 cm

17. An arc of length 22 cm subtends an angle at the centre of a circle of radius 42 cm. The angle subtended by the arc at the centre is: 1

(a) 15° **(b)** 60° **(c)** 30° **(d)** 150°

18. The radii of two solid spheres are in the ratio 2 : 3. The ratio of their surface areas is: 1

(a) 2 : 3 **(b)** 8 : 27 **(c)** 4 : 9 **(d)** 8 : 9

19. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 with frequencies 17, 21, 24, 17, 6, 4, 11 respectively, The mode of the data is: 1

(a) 68 **(b)** 60 **(c)** 63 **(d)** 70

20. A bag contains 11 red and 5 blue balls. 9 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{21}{25}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{19}{25}$

21. A number ends with exactly 2 zeros if its prime factorisation (other factor 3) is: 1

(a) $2^3 \times 5^2 \times 3$ **(b)** $2^3 \times 5^1 \times 3$ **(c)** $3^2 \times 7$ **(d)** $2^1 \times 5^2 \times 3$

22. If the zeroes of $x^2 - 10x + k$ differ by 4, then k is: 1

(a) 29 **(b)** 21 **(c)** 84 **(d)** 42

23. If $5x + 3y = 47$ and $3x + 5y = 41$, then $x - y =$ 1

(a) 6 **(b)** 44 **(c)** 3 **(d)** $\frac{3}{4}$

24. If α, β are roots of $x^2 - 5x + 6 = 0$, then $\dfrac{1}{\alpha} + \dfrac{1}{\beta} =$ 1

(a) 30 **(b)** $\frac{5}{6}$ **(c)** $-\frac{5}{6}$ **(d)** $\frac{6}{5}$

25. How many terms of the AP $20, 16, 12, 8, \ldots$ give sum 0? 1

(a) 12 **(b)** 11 **(c)** 22 **(d)** 10

26. A square has diagonal $5\sqrt{2}$ cm. Its area (cm²) is: 1

(a) 25 **(b)** 50 **(c)** 10 **(d)** 100

27. The three points $(0, 0)$, $(8, 0)$ and $(3, 7)$ enclose area: 1

(a) 56 **(b)** 14 **(c)** 42 **(d)** 28

28. The value of $\tan 10^\circ \tan 20^\circ \tan 70^\circ \tan 80^\circ$ is: 1

(a) 1 **(b)** 2 **(c)** $\tan 45^\circ + 1$ **(d)** 0

29. Two points on level ground lie on opposite sides of a tower of height 9 m, in line with its foot. The angles of elevation of the top of the tower from the two points are 45° and 60°. The distance between the two points is: 1

(a) 12 m **(b)** $9 - 3\sqrt{3}$ m **(c)** $9 + 9\sqrt{3}$ m **(d)** $9 + 3\sqrt{3}$ m

30. A point P is 85 cm from the centre O of a circle of radius 77 cm. The length of the tangent drawn from P to the circle is: 1

(a) 36 cm **(b)** 8 cm **(c)** 1296 cm **(d)** 162 cm

31. A square is inscribed in a circle of radius 35 cm. The area of the circle outside the square is: 1

(a) 2625 cm² **(b)** 3850 cm² **(c)** 2450 cm² **(d)** 1400 cm²

32. The radii of two solid spheres are in the ratio 2 : 3. The ratio of their volumes is: 1

(a) 8 : 27 **(b)** 4 : 9 **(c)** 16 : 27 **(d)** 2 : 3

33. For a distribution with median class 20–30, $N = 50$, $cf = 18$ and $f = 14$, the median is: 1

(a) 22.5 **(b)** 28 **(c)** 25 **(d)** 24

34. A box contains 70 tickets numbered 1 to 70. One ticket is drawn at random. The probability that the number drawn is a prime number is: 1

(a) $\frac{51}{70}$ **(b)** $\frac{9}{35}$ **(c)** $\frac{2}{7}$ **(d)** $\frac{19}{70}$

35. The LCM of 36, 48 and 72 is: 1

(a) 144 **(b)** 1728 **(c)** 5184 **(d)** 72

36. If $x = 2$ is a zero of $x^2 - 9x + 14$, the other zero is: 1

(a) -7 **(b)** 7 **(c)** 11 **(d)** 12

37. The pair $kx + 3y = k - 3$ and $12x + ky = k$ has no solution when $k =$ 1

(a) 3 **(b)** -6 **(c)** 6 **(d)** 12

38. If the roots of $x^2 - 9x + 14 = 0$ are α, β, then $\alpha^2 + \beta^2 =$ 1

(a) 53 **(b)** 109 **(c)** 81 **(d)** 67

39. If the 5th term of an AP is 11 and the 11th term is 5, then the 16th term is: 1

(a) 16 **(b)** 0 **(c)** 1 **(d)** 55

40. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 2 : 3$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 4: 1

(a) 29 **(b)** 21 **(c)** 9 **(d)** 25

41. **Assertion (A):** The area of a sector of a circle of radius 7 cm and angle 90° is $38.5\ \text{cm}^2$ (use $\pi = \frac{22}{7}$). 1

Reason (R): The area of a sector of angle θ is $\dfrac{\theta}{360^\circ} \times \pi r^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** The total surface area of a solid hemisphere of radius r is $3\pi r^2$. 1

Reason (R): The total surface area of a solid hemisphere is its curved surface area $2\pi r^2$ plus the area πr^2 of its flat circular face.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** For the class interval 10-20, the class mark is 15. 1

Reason (R): The class mark of a class interval is $\dfrac{\text{upper limit} + \text{lower limit}}{2}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** The probability of getting a number greater than 6 when a die is thrown once is 0. 1

Reason (R): The probability of an impossible event is 0.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** The number $5 - 2\sqrt{3}$ is irrational. 1
Reason (R): The sum of a rational number and an irrational number is irrational.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** The sum of the zeros of $2x^2 - 8x + 6$ is 4. 1
Reason (R): For $ax^2 + bx + c$ with zeros α, β, the sum of the zeros is $\frac{c}{a}$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** The pair $x + 2y = 4$, $2x + 4y = 12$ has no solution. 1
Reason (R): Two lines with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always intersect at exactly one point.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** The equation $x^2 - 6x + 9 = 0$ has two equal real roots. 1
Reason (R): A quadratic equation always has two distinct real roots.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** The 10th term of the AP $2, 5, 8, \ldots$ is 29. 1
Reason (R): The nth term of an AP with first term a and common difference d is $a_n = a + nd$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** In a triangle ABC right-angled at B, with $AB = 6$ cm and $BC = 8$ cm, the hypotenuse $AC = 10$ cm. 1
Reason (R): The Pythagoras theorem holds only for isosceles right-angled triangles.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**